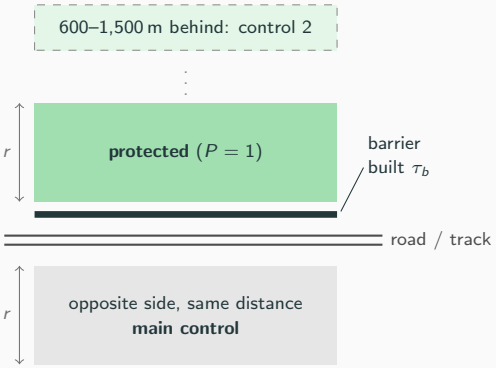
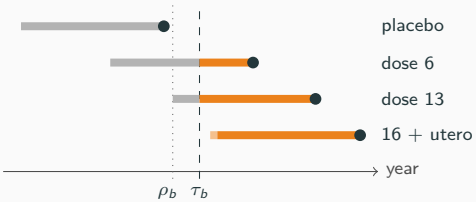
The design in one picture

Where: one corridor per barrier



When: birth cohorts around τ_b

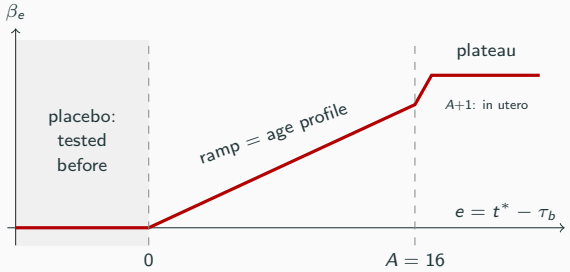


Bar: birth to test (•, age 16). Brown: years lived with the barrier.

Design: difference-in-differences across birth cohorts *within* each barrier. Children from protected cells against children from the opposite side of the same road, cohorts tested before against cohorts tested after. The mother's cell in ρ_b, before the project, decides P.

The prediction: zero, ramp, plateau

If home noise harms scores cumulatively



- $e < 0$: no exposure, $\beta_e = 0$
- Each later cohort adds one year, at a younger age: the step $\beta_e - \beta_{e-1}$ is the effect of exposure at age $A - e$
- $\beta_{A+1} - \beta_A$: exposure in utero
- Cohorts born after the barrier have the same dose: flat

Other stories draw other pictures



Pre-trend:
sorting, anticipation



Step, no ramp:
noise around the test



No plateau:
trends set off by the road project

Who counts as exposed: the cell before the project

Threat: families move in, move out or stay *because of* the barrier, and road projects are announced years ahead. Assigning by current address would build sorting into the ramp.

Assignment

- **Reference cell:** the mother's 100 m cell on 31 December of ρ_b, before the project could affect where families live
- ρ_b = year before the project was announced; with no project dates yet, $\rho_b = \tau_b - 3$ (robustness: $\tau_b - 1$, $\tau_b - 5$; later dates should give smaller estimates if anticipation matters)
- Through the mother, children born after τ_b get a cell by the same rule as their older siblings
- Children are followed after moves: an **intention-to-treat** effect. Moving becomes a mechanism, not a threat

Not: stayers only (as Lindgren 2021), which selects on a response to treatment; or dwellings (as Moretti and Wheeler 2025), which mixes the effect with who moves in and out.

Estimation: one stack per barrier

$$S_i = \sum_{e \neq -1} \beta_e \, \mathbf{1}\{e_i = e\} \, P_i + \gamma_{b,P_i,r(i)} + \eta_{b,e_i} + X'_{f(i)}\phi + \varepsilon_i$$

- S_i: grade-9 merit value as a percentile within the test year (grading changed over time)
- $\gamma_{b,P,r}$: each side $\times$ distance band, per barrier
- $\eta_{b,e}$: barrier $\times$ test year. Absorbs traffic, local shocks and the road project as far as it is the same at all distances. Robustness: $\eta_{b,r,e}$, same distance band only
- $X_{f(i)}$: parental education, income, immigrant background at ρ_b
- No school FE (school chosen after τ_b); robustness: *assigned* school $\times$ test year
- Event time and dose from the *intended* test year $t^* = a_i + 16$, so repeating a grade does not change them
- Each barrier is compared only with its own cells; cells protected by another barrier are dropped (Cengiz et al. 2019)
- Weighted by protected children, event study on barriers with a full window (built by about 2007) (Wing et al. 2024)
- Clustered by connected corridors; wild cluster bootstrap and randomisation inference
- **Headline:** effect per year of assigned dose, once pre-trends are flat and the ramp is roughly linear
- Cross-checks: Callaway and Sant'Anna (2021), Sun and Abraham (2021), Borusyak et al. (2024)

Is it the noise?

Road barriers come with a road project and lower near-road air pollution (Baldauf et al. 2008; Greenhill 2026). Three tests separate noise from the bundle:

Test	If it is noise	If it is the project or air pollution
Rail against road	Similar ramps: rail traffic is mostly electric, so a rail barrier changes little but noise	Ramp for road barriers only
Threshold, fixed in advance: L^0 above WHO guideline (53 dB road, 54 dB rail)	Ramp above the threshold, none below	Ramp below the threshold too, among cells as close to the road (same project) but quiet to begin with
Intensity: predicted dB the barrier removes at each cell	Larger ramp where the barrier removes more dB	Same ramp whatever the dB cut

Threshold and intensity compare cells at the same distance from the road: the project's effects also fade with distance.

Backup

Backup: overview

Slide	What it covers	Backs up
Threats and answers →	Each threat, the design answer and the evidence we show	All
Distance and comparison groups →	The three comparison groups; effect by 50 m bin; balance before the barrier	Design
Mobility and per year lived →	Moving as a mechanism; 2SLS on years actually lived	Assignment
Dose and age profile →	Parametric effect per year of dose; kinks by age; duration against intensity	Prediction, estimation
Other estimands and checks →	Siblings, residents at the test, heterogeneity, placebo checks	Threats
Literature →	What we take from each study and what we do differently	Contribution
References →		

Backup: threats and answers

Threat	Design answer	Evidence we show
Families sort or anticipate	Reference cell at ρ_b, via the mother; ITT	Mobility event study; births by side; predicted-score placebo
Road project comes with the barrier	Opposite side of the same road; barrier $\times$ year FE	Threshold, intensity, plateau; barrier $\times$ band $\times$ year FE
Air pollution moves with noise	Rail against road	NO_2 by side, if a grid exists
Staggered timing, bad comparisons	Stacking, own cells only, corrective weights	Callaway–Sant'Anna, Sun–Abraham, Borusyak et al.
Cells misclassified at the zone edge	Drop edge cells (100 m grid)	Estimates with them included
Comparison cells insulated	— (bias toward zero)	Insulation records, if they exist
Grading changes	Percentile within test year	Raw merit value, national tests
Sides feed different schools; grading by school	Assigned school $\times$ year FE (not actual school: post-treatment)	School overlap of the two sides; actual school $\times$ year FE as decomposition

◂ Overview

Backup: distance profile and comparison groups

- **Comparison groups:** (1) opposite side, same distance (main); (2) 600–1,500 m behind the barrier; (3) not-yet-treated corridors (robustness). Not never-treated places alone: places eligible but never shielded may have low priority for reasons that also shape their trends
- **Caveat for (1):** the unshielded side may lack a barrier because it has few dwellings or got insulation, which we check
- **Distance profile:** the effect in 50 m bins of distance from the barrier up to 1,500 m, each against the opposite side at the same distance. Shows how far the protection reaches, and whether the 600–1,500 m band is unaffected; if not, the band moves out
- **Balance:** for cohorts tested before the barrier, scores should vary with distance in the same way on both sides (Berkhout et al. 2025)

◂ Overview

Backup: mobility and the effect per year lived

$$M_{ft} = \sum_{k \neq -1} \mu_k \, \mathbf{1}\{t - \tau_b = k\} \, P_f + \gamma_{b,P_f,r(f)} + \eta_{b,t} + X_f'\phi + u_{ft}$$

- **Mobility as mechanism:** hazard of leaving the reference cell. Moves between ρ_b and τ_b indicate anticipation
- **Per year lived:** 2SLS with years actually lived behind a barrier, instrumented by the assigned dose: an average causal response among compliers (Angrist and Imbens 1995)
- A *LATE* if the barrier does not change moving or schools, else a *rescaled ITT*; bounds that allow a direct effect of assignment (Conley et al. 2012)
- Compliers described by household characteristics at ρ_b (Abadie 2003)

◂ Overview

Backup: effect per year of dose and age profile

$$S_i = \beta\, D_i P_i + \beta^{\text{utero}}\, \mathbf{1}\{e_i > A\}\, P_i + \beta^{\text{pre}}\, \mathbf{1}\{e_i < 0\}\, e_i\, P_i + \gamma_{b,P_i,r(i)} + \eta_{b,e_i} + X'_{f(i)}\phi + \varepsilon_i$$

$$D_i = \min\{\max(0,\ e_i),\ A\}$$

- β: effect per year of assigned dose; β^{utero}: added effect of exposure in pregnancy; β^{pre}: pre-trend test
- Age profile: slope allowed to differ by the age of the added year, kinks at 3, 7 and 13
- Not a regression on D_i alone: cohorts tested before would pool with comparison cells and the placebo would be lost; a continuous dose with staggered timing needs stronger parallel trends (Callaway et al. 2024)
- Duration against intensity: years of exposure holding dB-years $D_i \Delta L$ fixed (Hollingsworth et al. 2025); permutation of ΔL within corridors

◂ Overview

Backup: other estimands and checks

Other estimands

- **Siblings:** household fixed effects with birth order $\times$ sex fixed effects (Black et al. 2005); selected toward large families that stay, so robustness only
- **Residents at the test:** the dwelling-based estimand; its gap to the ITT measures sorting; compositional-change test (Sant'Anna and Xu 2025)

Heterogeneity, fixed in advance

- Sex; parental education, income, immigrant background; housing type
- Wind direction: heterogeneity by side only, since it carries pollution too

Checks

- Predicted score from household characteristics at ρ_b as a placebo outcome
- Births after ρ_b: number and mothers' characteristics by side
- τ_b shifted back five years
- Off-age testing by P_i after τ_b; children tested on time only
- Dated against undated barriers: location, length, road type, neighbourhood
- Birth outcomes for cohorts born after τ_b (access to be confirmed)

◂ Overview

Backup: how this relates to the literature

Study	We take	We do differently
Lindgren (2021): Swedish noise mitigation and health	The setting	Keep movers; same-road controls, not never-treated
Moretti and Wheeler (2025): barriers and house prices	The 600–1,500 m band	Follow people, not dwellings; no unbuilt barriers in Sweden
Greenhill (2026): noise and infant health	Noise and air pollution move together	Rail as a near noise-only treatment
Grönqvist et al. (2020): lead, Swedish cohorts	Cohort × exposure, threshold fixed in advance, percentile outcome	Place-specific timing; dose from years lived
Hollingsworth et al. (2025); Berkhout et al. (2025)	Duration against intensity; age profile; distance cut-off from the data	Noise, not lead

What we add: a cohort exposure design for noise and school achievement, with assignment before the barrier and a within-corridor comparison that nets out the road project.

◂ Overview

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